



جامعة الملك عبد الله
للعلوم والتقنية
King Abdullah University of
Science and Technology

أكاديمية كاوست
KAUST ACADEMY



AI Summer School 2026

June 28 – August 20, 2026
King Khalid University (KKU)
KAUST Academy

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Welcome to **AI Summer School 2026!**

An intensive, hands-on journey into frontier AI research and practice.



2023



2024



2025

KAUST Academy: Mission + Who's behind it



***Prof. Sultan
Albarakati***

***Associate Vice
President, Saudi
Talent Development,
KAUST***



***Prof. Naeemullah
Khan***

***Deputy Director,
Instructional
Associate
Professor, KAUST***



***Mr. Abdulrahim
Alghamdi***

***KAUST Academy
Manager.
Literally HIM***



Ms. Oroba Al Nasr

***Promotions &
Outreach Manager,
Specializations Prog
rams Manager***

Meet your morning instructors



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Dr. Mohammed Ahmed
Week 1



Dr. Muhammad Mubashar
Week 2-3



Mr. Abdallah Hammad
Week 4-5



Dr. Ankuj Arora
Week 6



Prof. Tanveer Hussain
Week 7



Dr. Salman Khan
Week 8

Meet your afternoon instructors



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Science and Technology



Prof. Amr Abdel-Hamid
Week 1



Poor Tas
Week 2



Dr. Ankuj Arora
Week 3



Dr. Muhammad Farooq
Week 4



Dr. Sara Durini
Week 5



Mr. Abdulhadi Zubailah
Week 6



Dr. Salah Abdeljabar
Week 7



Poor TAs
Week 8

Meet your TAs



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Ali Habibullah
The GOAT



Yazan Alshuaibi
Ali's Top 1% hater



Mohammad
Alshiekh



Mohamed
Eltayeb



Lama Ayash



Muhannad
Alhumaidi



Jana
Shata



Abdalla
Abdelrahman



Anas
Aldadi



Hussain
Alyafei



Hassan
Mohammed

Learning Journey Recap – Stage 2 (ML & DL)

Classic ML:

regression,
classification,
ensembles.

Optimization: SGD,
momentum, Adam;
regularization tactics.

Deep Learning

Basics: perceptron,
multilayer nets,
activations.

Tooling: hands-on
with scikit-learn &
PyTorch mini-labs.



Learning Journey Recap – Stage 3 (Computer Vision)

CNN & Transfer Learning:

convolutional layer,
residual learning,
EfficientNet.

Segmentation: Unet
& MaskRCNN

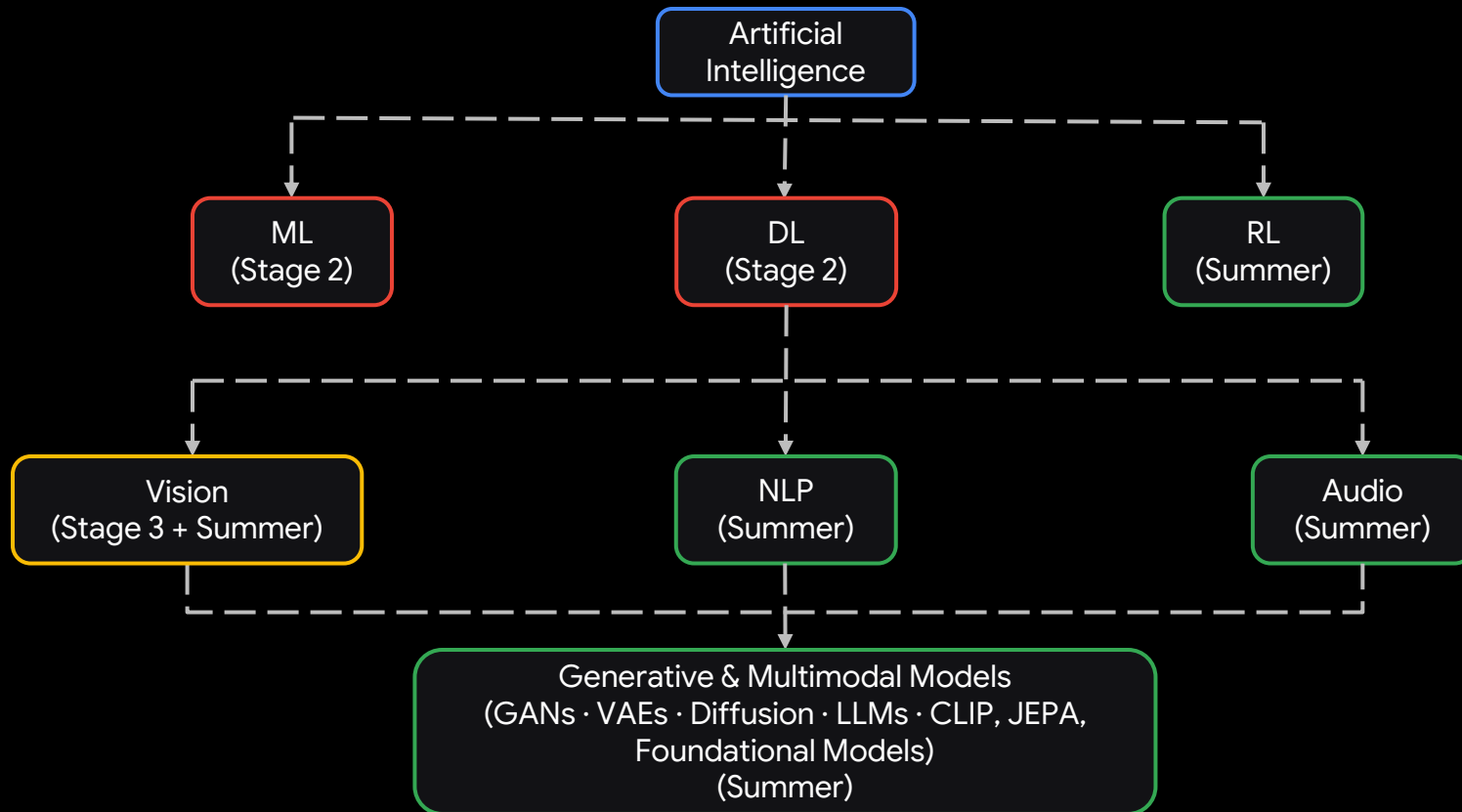
Practical Deep

Learning: data handling,
augmentation,
regularization.

Object Detection:

RCNN, FastRCNN,
FasterRCNN, YOLO

AI Landscape Overview



Summer Content



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Week	Topics	Instructor
1	CNNs, RNNs, Transformers, VIT, Videos	Dr. Mohammed Farouk Ahmed
2	SSL, CL, JEPA, World Models, Foundational Models	Dr. Muhammad Mubashar
3	AEs, VAEs, GANs, Diffusion, Stable Diffusion	Dr. Muhammad Mubashar
4	RL, Value Based, Stochastic Policy, Actor Critic	Mr. Abdallah Hammad
5	CC, Reward problems, RL in real world, RL Frontiers	Mr. Abdallah Hammad
6	Introduction to Natural Language Processing	Dr. Ankuj Arora
7	Transformers & LLMs, Reasoning, Quantization	Prof. Tanveer Hussain
8	Audio & Agents	Dr. Salman Khan

Afternoon Skills Track



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Week	Topics	Instructor
1	AI & IoT	Prof. Amr Abdel-Hamid
2	General Topics	TAs
3	MLOps	Dr. Ankuj Arora
4	Building Reliable Agents	Dr. Muhammad Farooq
5	Scientific Writing	Dr. Sara Durini
6	Game Development	Mr. Abdulhadi Zubailah
7	Communication & Visualization	Dr. Salah Abdeljabar
8	Project Work & Wrap-up	TAs

A Day at Summer School

Time	Activity	Format
9:00 – 10:30 AM	Morning Lecture	Technical track
11:00 – 12:30 PM	Lab Session	Hands-on coding
12:30 – 2:00 PM	Lunch Break	—
2:00 – 3:30 PM	Afternoon Session	Skills track
3:30 – 5:00 PM	Project Work	Team project
Fri & Sat	Weekend Break	—



Expected Learning Outcomes

- Understand and evaluate recent AI research publications.
- Build diverse models across generative, vision, RL, and NLP domains.
- Implement and extend methods from research papers.
- Gain proficiency with advanced AI tools (e.g., PyTorch, Hugging Face).
- Apply MLOps practices to deploy, scale, and monitor models in production.
- Build reliable AI agents and retrieval-augmented (RAG) systems.
- Communicate and visualize results effectively for technical and general audiences.
- Collaborate effectively in teams on an AI project.
- Write and publish papers in top AI conferences.

Website



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